ChenYang Wang

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EDUCATION

Hainan University **Sep 2019 - Jun 2023**

School of Computer Science and Technology

- Bachelor of Computer Science and Technology
- Thesis: "Machine Learning-Based DDoS Attack Detection and Research"
- Advisor: Professor Jieren Cheng
- Average Score: 87/100

RESEARCH INTERESTS

- AI for Security Vulnerability Discovery，AI for CTF
- LLM/Agent Security in Automated Systems

ACADEMIC PAPER

- S. Wu*, **C. Wang***, W. Yan, W. Qu, P. Chen, J. Hou, S. Li, Z. Kang. (2026). "Lowering the Barrier, Not the Threat: LLM-Assisted Web Bots in ATO-Relevant Workflows" (**Submitted to IEEE Transactions on Dependable and Secure Computing**)
- **C. Wang**, Y. Shi, Z. Kang. (2027). "PAC-Graph: Tracing Component-Level Contamination Propagation in LLM Agents" **(In preparation. Target venue: AAAI 2027)**

RESEARCH EXPERIENCE

Research Assistant - "Lowering the Barrier, Not the Threat: LLM-Assisted Web Bots in ATO-Relevant Workflows" **May 2025 - Present**

Advisor: Associate Researcher Zifeng Kang(BUPT) | Co-advisor: Associate Professor Shujiang Wu(BUAA)

- **Objective:** Conducted a systematic empirical study evaluating whether LLM-driven web bots lower the barrier for non-expert attackers to perform account takeover (ATO) attacks against real-world websites.
- **Methodology:** Collected and benchmarked 20+ open-source web bot tools across three categories (Traditional Tools, LLM-assisted Coding, and Autonomous Agents). Designed a three-stage experimental framework covering capability assessment under zero-defense conditions, attacker cost-benefit analysis, and defense effectiveness evaluation against signal-based anti-bot mechanisms.
- **Contribution:** Led literature survey, experiment design and implementation, data analysis, and manuscript writing.
- **Outcome: Paper co-first authored and submitted to IEEE TDSC.**

Research Project - "PAC-Graph: Tracing Component-Level Contamination Propagation in LLM Agents"

March 2026 - Present

Advisor: Associate Researcher Zifeng Kang(BUPT)

- **Objective:** Developed a component-level evaluation framework to measure how adversarial information propagates through planner, policy, tool, and memory components in LLM agent systems.
- **Methodology:** Methodology: Designed PAC-Graph, a component-level contamination measurement framework using controlled prompt injection as an experimental probe. Conducted large-scale evaluations on AgentDojo with models to analyze propagation behavior across planner, policy, and tool/data components.
- **Contribution:** Led literature review, threat model formulation, framework design, experimental implementation, statistical analysis, and manuscript preparation.
- **Outcome: Manuscript in preparation. Target venue: AAAI 2027.**

Undergraduate Thesis Research - Machine Learning-Based DDoS Attack Detection and Research

Jan 2023 - May 2023

Advisor: Professor Jieren Cheng

- **Objective:** Investigated machine learning approaches for detecting DDoS attack traffic.

- **Methodology:** Implemented and compared multiple ML models (e.g., Random Forest, SVM, Neural Networks) for classifying DDoS traffic on benchmark datasets.
- **Outcome:** Completed thesis evaluating detection accuracy and efficiency, demonstrating the effectiveness of ensemble methods for real-time DDoS identification.
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Frontend Developer - Blockchain-Based Evidence Preservation System, Hainan University

Jun 2021 - Apr 2022

Advisor: Professor Chunjie Cao

- **Objective:** Developed a consortium blockchain-based digital evidence preservation system for judicial departments (government-funded project).
- **Contribution:** Built user-facing frontend pages for evidence submission and data querying using Vue.js.

Project Member (Competition) - Deep Learning-Based Fatigue Driving Detection System, Hainan University

Sep 2020 - Mar 2021

Advisor: Professor Jieren Cheng

- **Objective:** Built a real-time driver fatigue detection system integrating face localization (LFPN/PyramidBox), eye/mouth fatigue monitoring (Haar-like classifier), and emotion recognition (VGG16).
- **Contribution:** Developed the face localization module, conducted functional testing, and participated in project proposal writing and competition defense.
- **Outcome: Awarded First Prize and Practical Value Award (National Level)** at the 15th Pan-Pearl River Delta University Computer Works Competition.

Member - HnuSec Cybersecurity Lab, Hainan University **2021 - 2023**

- Trained and competed in Capture The Flag (CTF) competitions as a member of the university cybersecurity team.

WORK EXPERIENCE

Quantitative Business Support (On-site) - Huabao Securities, Shanghai **Jun 2025 - Present**

- Maintained quantitative trading test environments and developed Python-based data processing and automated reporting tools with GUI for non-technical users.

Security Engineer - Shanghai Chuangqi Tianxia Technology Co., Ltd. **Jun 2024 - May 2025**

- Developed an intrusion detection system (IDS) rule engine module; designed detection rules based on HTTP traffic features and anomalous behavior patterns, reducing false positive rate from 15% to 12%. Participated in backend database optimization using Redis, improving engine efficiency by 10%.
- Conducted penetration testing for 20+ financial and government institutions, covering black-box testing, vulnerability verification, and remediation reporting. Discovered multiple critical vulnerabilities including unauthorized access and arbitrary file upload.
- Performed source code security auditing on Python and C++ projects using Fortify, identifying potential security risks and providing remediation recommendations.
- Participated in vulnerability discovery projects assigned by Hubei Provincial Communications Administration, identifying multiple critical vulnerabilities across municipal and enterprise systems.
- Contributed to the team's recognition as a "2024 Outstanding Collective for Cybersecurity and Data Security in Shanghai" by the Shanghai Municipal authorities.

HONORS & AWARDS

2020

- First Prize & Practical Value Award (**National**), The 15th Pan-Pearl River Delta University Computer Works Competition
- **National** Encouragement Scholarship (CNY 5,000)
- Second Prize (**Province**), The 6th China International "Internet+" College Students Innovation and Entrepreneurship Competition
- Merit Student Award (Hainan **University**)

2021

- Second Prize (**Province**), China Collegiate Computing Contest - Network Attack and Defense Competition (CTF)

SKILLS

- **Languages:** Mandarin (Native), English (CET-6, preparing for IELTS)
- **Programming:** Python, JavaScript, SQL, LaTeX
- **Security Tools:** Wireshark, Burp Suite, Kali Linux, Fortify
- **Platforms & Tools:** Git, Docker, Linux
- **Interests:** CTF competitions, swimming